⇒ Bus Data — ✕ Later

⇒ Line Data ←

	St. Bus	End Bus	r	x	b
Line 1	1	2	r_{12}	x_{12}	$b/2$
Line 2	1	4	r_{14}	x_{14}	
line 3	2	3	r_{23}	x_{23}	
line 4	3	4	r_{34}	x_{34}	

⇒ Read Data

y_{12} y_{13} y_{23} y_{24}

$y_{14} = 0$ $n = 4$

for i = 1 , n
  for j = 1, n
    if (i = j)
      Y(i,j) = sum of all admittance connected
    else
      Y(i,j) = - y(i,j)
  end
end

YBus
100

n Buses

P_i , Q_i

$$I_i = \sum_{j=1}^{n} Y_{ij} V_j$$

$P_i = P_{geni} - P_{loadi}$ ⇒ real power Injected at bus i

$Q_i = Q_{geni} - Q_{loadi}$ ⇒ reactive power Injected at bus i

If $P_{geni} > P_{loadi} \Rightarrow P_i \Rightarrow$ (+ve) ⟸ Gen Bus

If $P_{geni} < P_{loadi} \Rightarrow P_i \Rightarrow$ (-ve) ⟸ Load Bus

$Q_i \Rightarrow$ +ve / -ve

$I_i \Rightarrow$ relationship with P_i and Q_i

$$S_i = V_i I_i^* = P_i + jQ_i$$

$$S_i^* = P_i - jQ_i = V_i^* I_i$$

$$\leftarrow I_i = \frac{P_i - jQ_i}{V_i^*}$$

Basic Power flow equations

$$\frac{P_i - jQ_i}{V_i^*} = \sum_{j=1}^{n} Y_{ij} V_j$$

10/09/26

Recap

① Real power balancing

② Reactive power balancing

HV Syst

modeling → transmiss

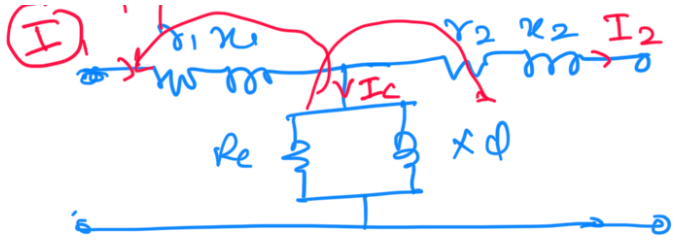
not dist. syst



⟸ approximate Equivalent circuit for transformer

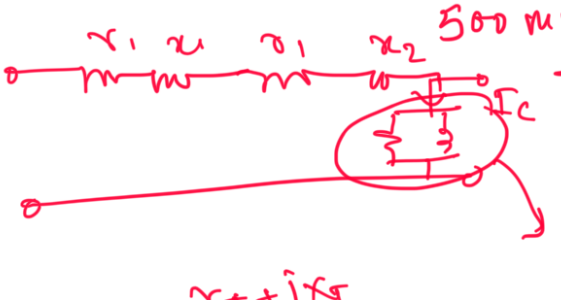


↳ at power transformer level → ✓ ok to approximate



← exact equivalent ckt.

on which grounds did we move to approx.



500 MVA, 11KV ← load current

I_c ⇒ core loss ← 2 to 6 Y. of load current

X

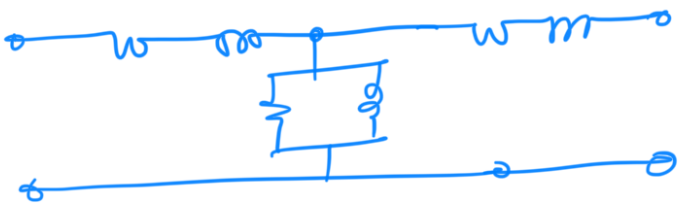
neglected



$x_T >> r_T$ ⇒

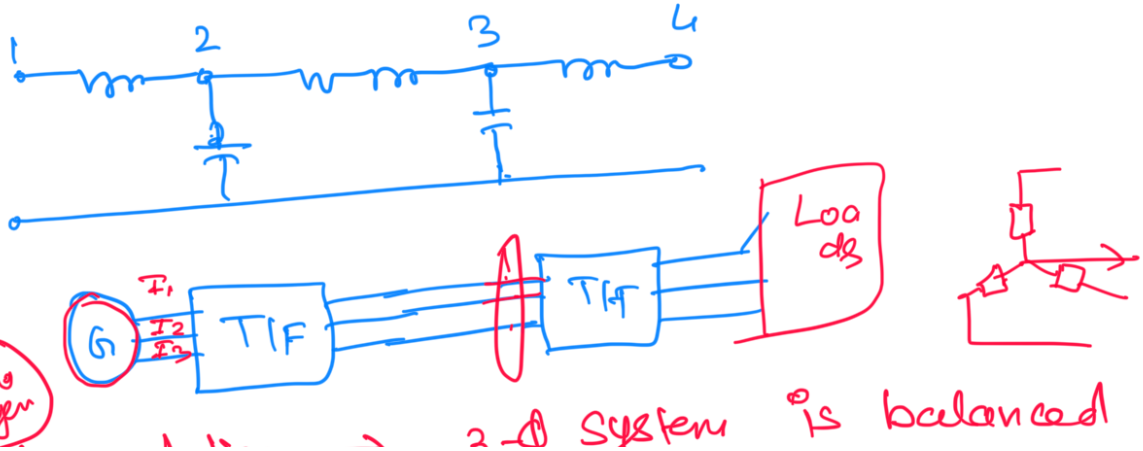
approx.

Low voltage level → 400V, 11KV below



Full model

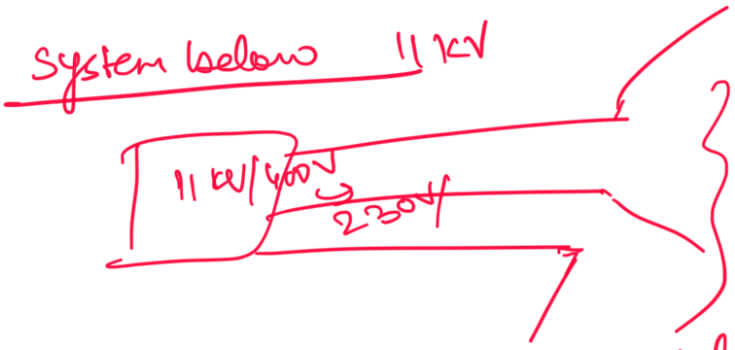
Load on a distribution transformer would vary throughout the day ← It would not be carrying full load current all the time.



3-Φ system is balanced

Assumption ⇒ 66KV | 400KV | 230KV

System below 11KV



Complete modeling of each component in 3-phases

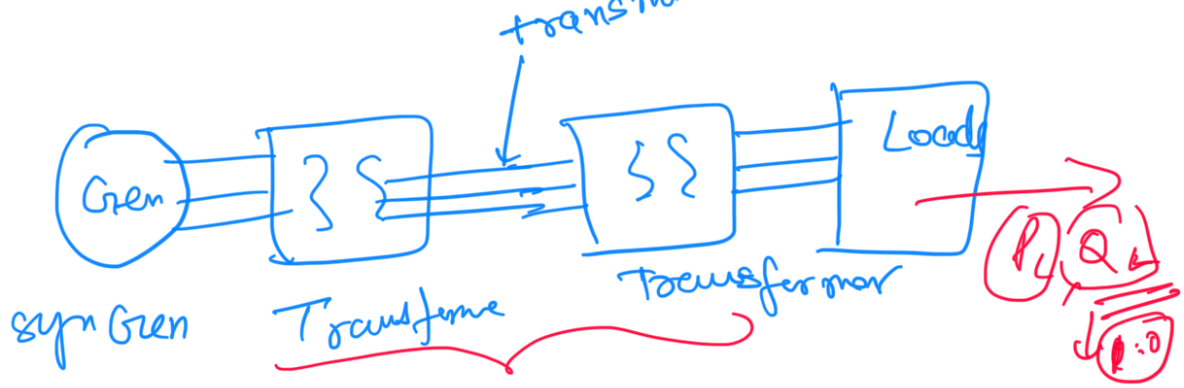
1 2 [ ]

$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$

Detailed modeling

$Y_{11} \Rightarrow 3\times3$

$= \begin{bmatrix} Y_{11RR} & Y_{11RY} & Y_{11RB} \\ Y_{11YR} & Y_{11YY} & Y_{11YB} \\ Y_{11BR} & Y_{11BY} & Y_{11BB} \end{bmatrix}$



Synchronous Gen ⟶ P_{gen}, Q_{gen}

↳ can provide active as well as reactive power and can fairly control them independent of each other

P_{gen} ⟵ Input flow

Q_{gen} ← f_y excitation

Transmission Line

Capacity of transmission line → MVA

Capacity of TIF → MVA/KVA



$S_{12_{max}} = 500\ MVA$

$S_{12} = \sqrt{P_{12}^2 + Q_{12}^2}$ → utilize

all attempts are there to utilize S_{12max} towards providing real power

Mostly → at transmission level ← P.F.

Power factor of loads should be as close to unity as possible

a) Real power load - gen balance

$$\sum_{i=1}^{NG} P_{Gi} = \sum_{j=1}^{n} P_{loadj} + P_{loss_{total}}$$

↳ frequency in the system is effected

b) Reactive power load gen balancing

$$\sum_{i=1}^{NR} Q_{Gi} = \sum_{j=1}^{n} Q_{loadj} + Q_{loss_{total}}$$

2.5 3.5

⇓

voltage at buses would be effected

assured ⇒ Three level of controls → fastest

gen. sense change in freq. and will respond ⇒ primary control → local

⇒ secondary control ⇒ 2nd

automatic generation control

⇓

co-ordinate control between various generators

⇓

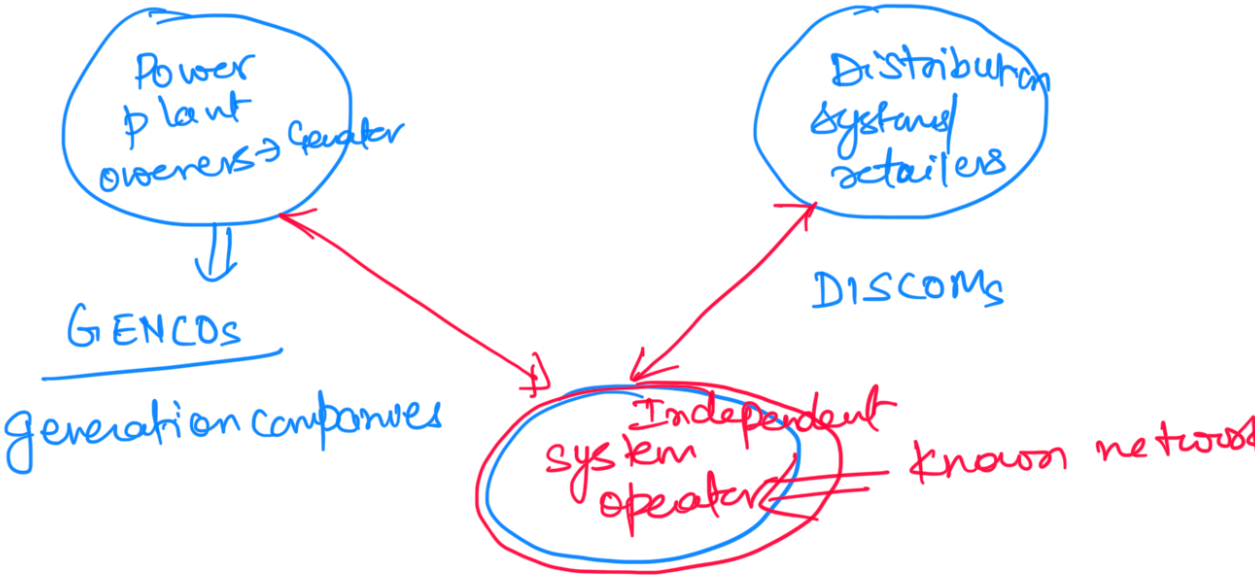
communication and co-ordination layer

⇒ tertiary control → slowest

Planning load-Gen balancing advance in day ahead fashion

⇓

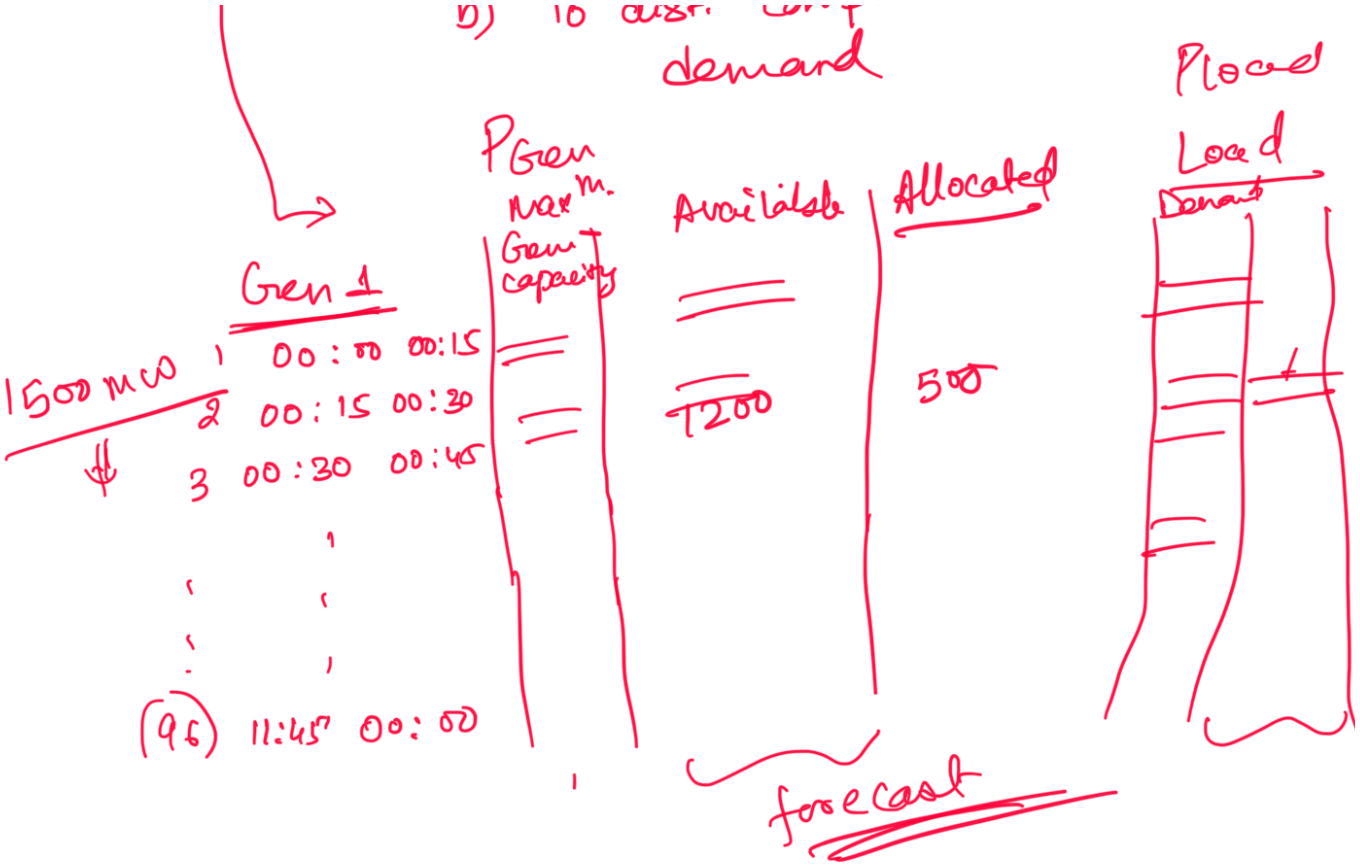
Done ?



⇒ one day in advance

a) Ask generators to provide their availability

b) [illegible] companies ← load



Total gen = Total load + Total loss + Reserves (min)

fast / slow

1. Tut - 1
2. Tut - 2
3. ✗ Bus → code